# 6.2 The Natural Base e

Algebra II 6.2 · Student sheet

Name _____ Period ____ Date _____

**I can…**

- define and use the natural base e.
- graph natural base functions and tell growth from decay.
- solve real-life problems with continuously compounded interest.

**Vocabulary** · fill in the missing word

**Natural base e** — an _____ number, $e \approx 2.71828$, the value $(1 + \frac{1}{n})^n$ approaches as n gets very large

**Natural base exponential function** — $y = ae^{rx}$: _____ when $a > 0$ and $r > 0$, decay when $a > 0$ and $r < 0$

**Continuously compounded interest** — $A =$ _____

## Part 1 · Compound more often

One dollar earns 100% interest for one year, compounded n times. Find $(1 + 1/n)^n$ to four decimal places.

| n | 1 | 2 | 4 | 12 | 365 | 8,760 |
|---|---|---|---|---|---|---|
| $(1 + 1/n)^n$ | | | | | | |

Now type $e^1$ on your calculator: _____ What do you notice?

## Part 2 · Notes and examples (fill in each note, then work the example)

**The natural base e:** $e \approx$ _____. It is the value $(1 + 1/n)^n$ approaches as n _____.

### Example 1

Simplify each expression.

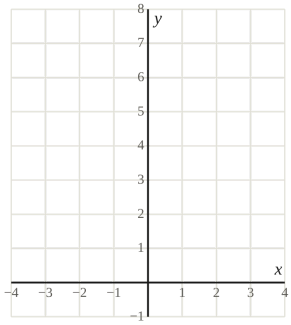
a. $e^3 \cdot e^5$ b. $\frac{16e^5}{4e^4}$ c. $(3e^{-4x})^2$

**Growth or decay:** $y = ae^{rx}$ $(a > 0)$ shows growth when _____ and decay when _____.

### Example 2

Tell whether each function represents exponential growth or decay. Then graph them.

a. $y = 3e^{2x}$  b. $y = e^{(-0.5x)}$




**Rewriting:** $e^{rt} = (e^r)^t$, so the growth factor is _____.

### Example 3

Rewrite $y = 2e^{(0.06t)}$ in the form $y = a(1 + r)^t$. Find the percent rate of change.

**Continuously compounded interest:** A = _____

### Example 4

You deposit $4,000 in an account that pays 5% annual interest compounded continuously. Find the balance after 6 years.

### Example 5

Find the balance for the same $4,000 at 5% for 6 years compounded quarterly. Which account earns more?

## Part 3 · Partner practice

### Partner 1

Simplify $(2e^{3x})^3$.

### Partner 2

You deposit $2,500 at 4% annual interest compounded continuously. Find the balance after 10 years.